Predicting Intoxication from Motion Data: A Machine Learning Analysis for JITAI Applications

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Abstract

Excessive alcohol use among college students remains a critical public health concern, and current prevention efforts often fail to intervene at the moment risk emerges. This work evaluates whether smartphone accelerometer data can predict intoxication states to support Just-in-Time Adaptive Interventions (JITAIs). Using a dataset of thirteen participants equipped with transdermal alcohol concentration (TAC) sensors and mobile motion tracking, we preprocess irregular sampling, address class imbalance, and engineer time- and frequency-domain gait features. We then benchmark linear models, ensemble methods, and neural networks. Results show that tree-based approaches (Random Forest, XGBoost) achieve the strongest performance (86.5% accuracy), surpassing linear models and multilayer perceptrons. Feature importance analysis reveals that impairment is best

detected through rhythmic stability and undampened impact forces rather than motion amplitude. These findings demonstrate the feasibility of motion-based intoxication inference and highlight its potential for delivering preventive, real-time interventions in high-risk environments.

1 Introduction

1.1 Motivation

Excessive alcohol consumption is a major yet preventable public health risk. In 2019, alcohol use accounted for an estimated 2.6 million deaths worldwide (World Health Organization, 2024). Heavy alcohol consumption is particularly prevalent on US college campuses, where the cultural and social environment reinforces the risk. (National Institute on Alcohol Abuse and Alcoholism, 2025). While traditional education-based interventions are well-intentioned, they often fail to provide help at the critical moment of decision-making. Just-in-Time-Adaptative-Interventions (JITAIs) offer a promising alternative by utilizing real-time smartphone data. Because alcohol physically impairs motor skills, these systems can track movement changes to identify intoxication states, delivering support precisely when it is most effective (Nahum-Shani et al., 2018). The following work focuses on delivering a preventive message shortly before a high-risk drinking episode in the context of a college binge-drinking event.

1.2 Machine Learning Problem Formulation

A key challenge in designing JITAIs is determining when to intervene. Failing to deliver a message when an individual is at risk can have severe safety implications. Conversely,intervening at inappropriate times can undermine efficacy by causing user annoyance or habituation (Serre et al., 2015). This justifies the need for JITAIs to be highly predictive and precise. We therefore formulate the following machine learning problem : How can machine learning models leverage real-time sensor data to accurately predict an individual's risk of heavy drinking, in order to trigger JITAIs only at moments when intervention is likely to be beneficial?

1.3 Data and Context of the Report

This study uses an open-source dataset made available by Killian et al.. In their experiment, the authors recruited thirteen college students who participated in a campus pub-crawl event. Each participant wore a transdermal alcohol content (TAC) ankle bracelet to measure alcohol intoxication levels and carried a smartphone that recorded motion data through its built-in accelerometer throughout the event.

1.4 Prior Work and Achieved Accuracy for this Problem

Prior work shows that mobile sensing can predict alcohol intoxication with varying accuracy depending largely on how the data is modeled. Early cross-person smartphone–gait models performed modestly, achieving roughly 56% accuracy, with Random Forest achieving the best performance among the evaluated models (Arnold et al., 2015). In contrast, within-person models performed substantially better : logistic regressions reached 90–93% accuracy and equivalent sensitivities (Suffoletto et al., 2020). More recent deep-learning models achieve RMSE values near 0.017 on Blood Alcohol Concentration (BAC) prediction (0.00–0.20 scale), corresponding to only ± 0.02 BAC units of average error, with Bi-LSTM and CNN models performing best (0.0167–0.0168) (Li et al., 2021).

In the original study by Killian et al. from which our data is derived, Random Forest achieved the highest accuracy (77.5%), outperforming the MLP (73.3%), CNN (74.3%), and SVM (74.7%) models. Our approach, however, diverges substantially from that of the authors—see Section 2—most notably in identifying limitations in the raw data, restructuring the dataset, and considering additional models.

2 Data Pre-processing

2.1 Raw Data

The data published by the authors is not immediately ready for analysis. The raw dataset is divided into two components:

1. **TAC readings**, from which the target intoxication labels are derived following the original paper, which defines $TAC > 0.08\ g/dl$ as intoxicated[1]; and
2. **Accelerometer recordings**, consisting of timestamps and tri-axial acceleration values—linear accelerations measured along three orthogonal axes corresponding to the device's x (medio-lateral), y (antero-posterior), and z (vertical) directions (see Figure 4)—from which gait and stability features are engineered during pre-processing.

2.2 Pre-processing : Handling inconsistent sampling rate

Although the original paper reports that accelerometer data were sampled at 40 Hz, the timestamps reveal highly irregular sampling intervals (see Figure 5). This is further confirmed by the examination of the inter-sample time intervals : while the median interval is consistent at 25 ms, gaps of several hours appear (Table 2).We corrected this discontinuity by upsampling to 1000 Hz using last-value imputation.

[1] This corresponds to the legal limit while driving in the context of Killian et al.

2.3 Pre-processing : Handling class imbalance

Because TAC is sampled only every 30 min compared to the faster accelerometer sampling, the naturally longer-lasting sober state is expected to dominate in the TAC readings as is confirmed by the data exploration (Figure 6). To correct this, we propagated TAC labels using forward-fill imputation, increasing the 'intoxicated' class representation from 24% to 36%.

2.4 Pre-processing : Generating features

Following the paper's two-tiered window approach, for every 10-second segment we further segmented the window down to 1-second segments. Aligning with standard time series classification, we engineered both :

1. **Time Domain:** Local sub-segment statistics to track signal evolution, supplemented by global window percentiles.
2. **Frequency Domain:** Spectral energy and centroid features calculated over the entire 10 s context.

The resulting dataset has 72521 rows and 153 features, each row corresponding to a window of 10 seconds (for a full list a features, see Table 3)

3 Results

To address our machine learning problem, we evaluated interpretable linear models (Logistic Regression, Ridge Classifier, and Support Vector Machine), ensemble methods (Random Forest, AdaBoost, XGBoost, Stacked Ensemble, and Voting Classifier), and a neural network (multilayer perceptron).

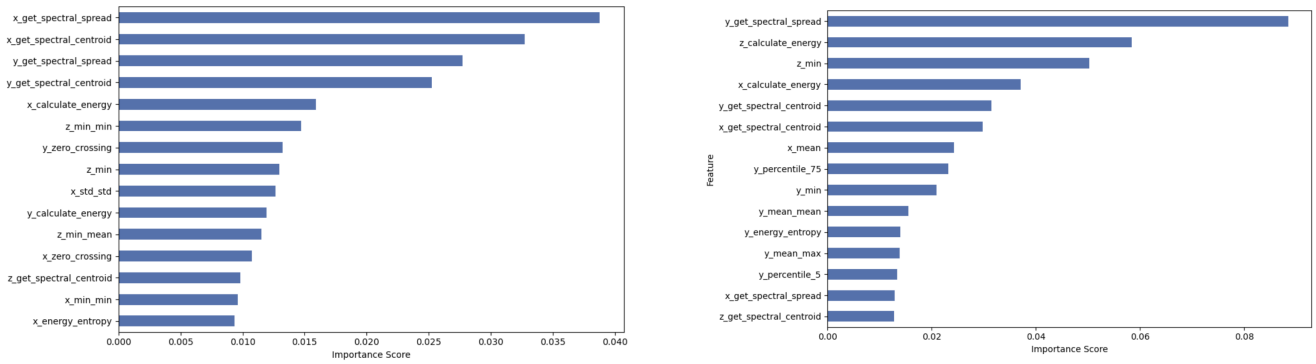
3.1 What the Model Sees : Feature-level insights into alcohol impairment

3.1.1 Detecting Impairment through Rhythmic Instability

The models challenge our intuition by asking not 'How much are the subjects moving?' but rather 'How chaotic is that movement?'. Indeed, our analysis shows that the models place far less emphasis on time-domain magnitude and instead focus on frequency-domain features, such as spectral spread, which have previously been shown to capture alcohol-induced disruptions in the periodicity of gait (Qi et al., 2025). This suggests the model has learned that the defining signature of intoxication is not how hard the subject moves, but their rhythmic stationarity. Our claim is further supported by the presence of zero-crossing features among the top predictors in the Random Forest model, which ask: "how often does the signal fluctuate around its equilibrium", reflecting oscillatory and unstable motion to distinguish intoxication states.

3.1.2 Detecting Impairment through Loss of Foot Impact Dampening

The feature importance analysis suggests that the XGBoost model's decision boundary is heavily influenced by a proxy for impact magnitude: vertical axis deceleration *z_min*. Effectively, the model asks, 'Did the subject catch their weight, or crash into the step?' to distinguish classes. This indicates that one of the most reliable discriminator of intoxication is not walking speed, but the presence of undampened heel strikes which result in extreme negative acceleration values distinct from the controlled landings of a sober gait(Wikipedia, 2025).[2].



(a) Random Forest

(b) XGBoost

Figure 1: Feature importance analysis for the top 15 predictors.

3.2 Why Complexity Matters : Connecting Feature Structure to Model Performance

Our results reveal a stark dichotomy in model performance, attributable to the step-like structure in the data resulting from pre-processing. Forward-fill imputation produces piecewise-constant feature trajectories with abrupt

[2]The point-biserial correlation ($p < 10^{-100}$) reveals a distinct physiological split: the mean $z_m invalue drops from a baseline of 7.48 \times 10^6$ in sober individuals to 2.24×10^6 in intoxicated subjects

transitions. Linear models and neural networks, which are biased toward learning smoothly varying functions, are therefore poorly matched to this structure. In contrast, tree-based models, which rely on recursive partitioning and naturally represent piecewise-constant functions, are well suited to the geometry of our data (Grinsztajn and Varoquaux, 2022).

3.2.1 Inductive Bias Limitations of Linear Models and MLPs

Linear models present the worst overall performance across all families of models. Logistic Regression, Ridge Classifier and Linear SVC correctly identify around 63% of the total instances -below the 64% accuracy achieved by a naive classifier always predicting non-intoxication -and tend to aggressively predict the minority class [3], leading to low specificity and precision, which both concentrate around 50%. The Linear SVM with transformed Random Fourier Features (RBF-approximated SVM) yields only marginal performance gains over the baseline linear SVM model. This is attributable to the nearly linear behaviour of the classifier, with an optimized kernel parameter of 0.01.

Our MLP model outperforms that of the original paper, correctly identifying 77.5% of the instances (vs 73.28%), and has a higher ability to discriminate between classes and to confidently predict positives, with a sober accuracy of 86.% (vs. 75.6%) and precision of 71.5% (vs. 59.8%). We attribute these discrepancies to differences in preprocessing and feature selection done in the original paper.

3.2.2 Superiority of Tree-Based Approaches

Ensemble methods consistently outperform linear models and the MLP. XGBoost and Random Forest yield the highest overall accuracy, of around 86.5%, and have a strong discriminative capacity, correctly identifying 88% of the non-intoxicated samples and 81-83% of the intoxicated ones. Their predictions are highly precise, with around 80% of their predicted positives being actual positives. Their consistent performance suggest the patterns are stable and not sensitive to their specific optimization method. This is also reflected in the minor marginal gains from combining both models with Stacking and Voting Classifiers. The lower performance of the AdaBoost, which reaches an overall accuracy of 74%, might be explained by a higher sensitivity to outliers.

Table 1: Model Performance in Test

Classifier	**Accuracy**	**Sober Accuracy**	**Precision**	**Recall**
Logistic Regression	0.6344	0.5204	0.4924	0.8403
Ridge Classifier	0.6342	0.5284	0.4922	0.8432
Linear SVC	0.6347	0.5194	0.4927	0.8429
Linear SVC + Fourier features	0.6569	0.5394	0.5109	0.869
Random Forest	0.865	0.8952	0.8106	0.8104
AdaBoost	0.7408	0.6825	0.5961	0.8462
XGBoost	0.8642	0.8815	0.7956	0.8331
MLP	0.7754	0.8643	0.7151	0.6144
Stacked Ensemble	0.8665	0.9016	0.82	0.8014
Voting Classifier	0.8665	0.8859	0.8014	0.8318

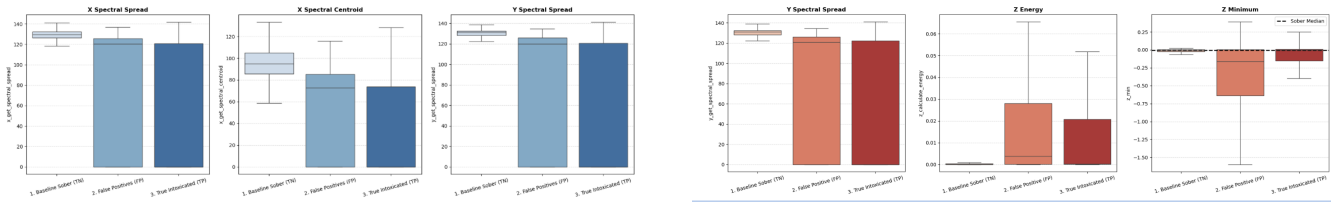
3.3 What Misclassifications Reveal: Detecting the Onset of Impairment

Perhaps the most persuasive argument for using this system in a JITAI setting comes from analyzing where the model "failed." To understand the operational limits of the model, we analyzed the feature distributions of the misclassified instances by our best-performing classifers XGboost and Random Forest.

3.3.1 False Positives: Hypersensitivity to Early Ataxia

The "False Positives" ($TAC < 0.08\ g/dl$) observed in our study appear to represent a hypersensitivity to early ataxia rather than random noise. Feature-level analysis demonstrates that these samples consistently share the physiological signature of intoxicated gait across the most influential predictors of each model. In both XGBoost and Random Forest, these 'sober' instances actually sit more on the stable end of the spectrum of movement, as the intoxicated samples. Their distribution likely reflects limited movement amplitude or postural rigidity—simply put, they are maintaining stability by not moving, rather than losing control.

[3] All models were trained using balanced class weights to reduce biased predictions toward the majority class. This explains a more aggressive behavior when predicting 'intoxicated' samples.



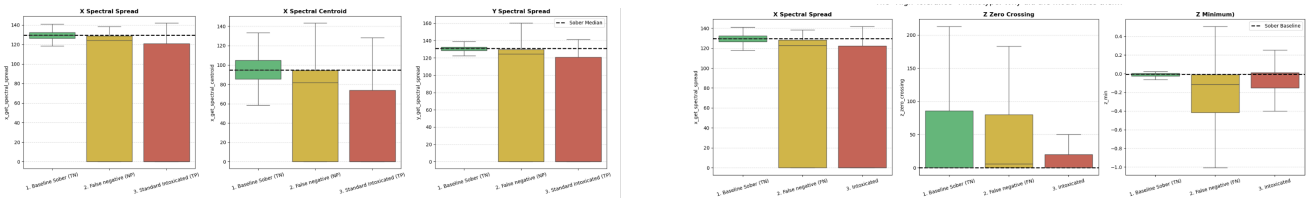
(a) Random Forest (b) XGBoost

Figure 2: False Positive analysis

3.3.2 False Negatives: The Hyperactive Phenotype

While the False Negative (FN) subjects exhibit feature distributions that clearly deviate from the Baseline Sober group, they do not align perfectly with the "standard" Intoxicated profile learned by the model. As seen in the **X Spectral Spread**, although the FN group aligns distributionally with the Intoxicated cases, they exhibit a higher median spread. This implies that these individuals maintained enough rhythmic inconsistency to appear sober to the model, contrasting with the lower spread of the True Positives. However, a crucial divergence appears in the vertical axis features. The correctly identified Intoxicated subjects (True Positives) exhibit low **Z Zero Crossings** and moderate **Z Minimum** values, indicative of a *hypo-active* or sedated gait strategy (sluggishness and reduced movement frequency).

In stark contrast, the False Negative group exhibits extremely high variance and magnitude in the Z-axis (peaking far above both Sober and Intoxicated groups). This suggests a phenotype of **"Hyperactive Instability"**—characterized by rapid corrective steps, stumbling, or erratic agitation. The model appears to have learned a decision boundary that associates intoxication primarily with gait suppression (hypo-activity). Consequently, when presented with the high-frequency, high-magnitude signals of the FN group, the model likely conflates this erratic energy with "active sobriety" (e.g., brisk walking), failing to recognize that this specific type of high-energy instability is also a manifestation of intoxication (see Figure **??**)



(a) Random Forest (b) XGBoost

Figure 3: False Negative analysis

3.4 Implications for JITAI

The observed misclassification patterns is not surprising given our approach: the models prioritize kinematic integrity over chemical concentration. By flagging early ataxia (False Positives) while missing the hyperactive phenotype (False Negatives), the system demonstrates that it is tracking functional behavior rather than strict legal thresholds.

For a JITAI system, this is not a flaw, but can be leveraged as an advantage. A safety intervention is most valuable when an individual's physical capabilities change, regardless of the specific TAC number. By triggering alerts based on the onset of gait patterns—whether sub-clinical or overt—the system delivers safety warnings based on actual behavioral vulnerability rather than a rigid metric.

4 Conclusion

Drawing on data from Killian et al., we train several machine learning models to predict risk of alcohol intoxication using smartphone accelerometer data from 13 college students during a binge-drinking event. Our results suggest that movement consistency —regular gait patterns and motor control primarily captured by frequency domain features —is more predictive than movement intensity —how much individuals move reflected by time domain features. Tree-based methods consistently outperform linear models and a MLP, which we attribute to the tabular and piece-wise constant nature of the data. Our two best performing models —a Random Forest and XGboost —achieve a 86.5% accuracy and demonstrate a strong class discrimination capacity. However, an examination

of misclassified instances highlights an important limitation: motor impairment alone can, in some cases, be an unreliable proxy for alcohol intoxication.

The study presents four main limitations. First, we did not account for inter-subject variability resulting from biological factors, like height, weight, and BMI, that influence the raw magnitude of motion data. Secondly, our models did not consider individual dependencies, ignoring that movement patterns are highly individual-specific. Thus, they looked for a "universal drunk gait" rather than identifying how a specific person's movement changed. Subject-wise normalization of sensor data with physiological traits and a sober baseline would account for these differences (Li et al., 2023). Thirdly, our 10-second window approach assumes window independence, disregarding the sequential nature of alcohol intoxication. Models such as LSTM could which track the progression of gait instablity could solve this limitation. Finally, the sample is highly context-specific, which hinders results generalizability.

5 Appendix

5.1 Raw Features Illustration

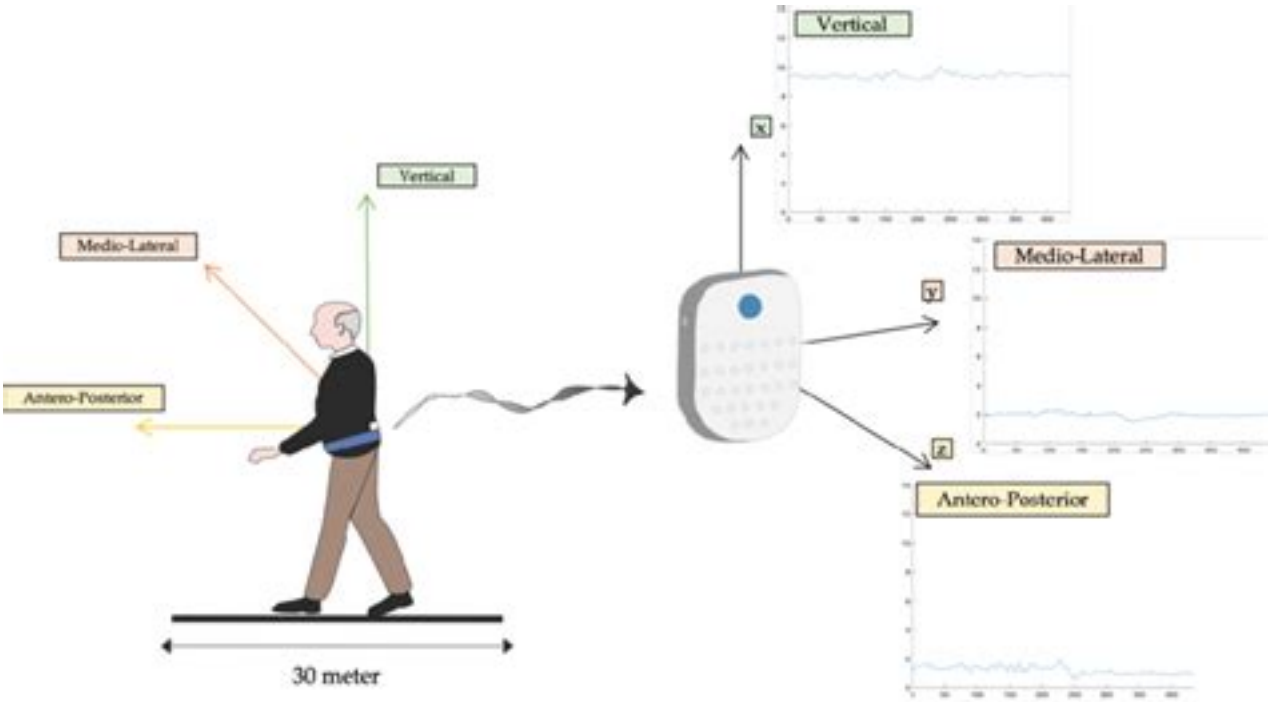


Figure 4: **Illustration of tri-axial acceleration**

5.2 Pre-processing Visualizations

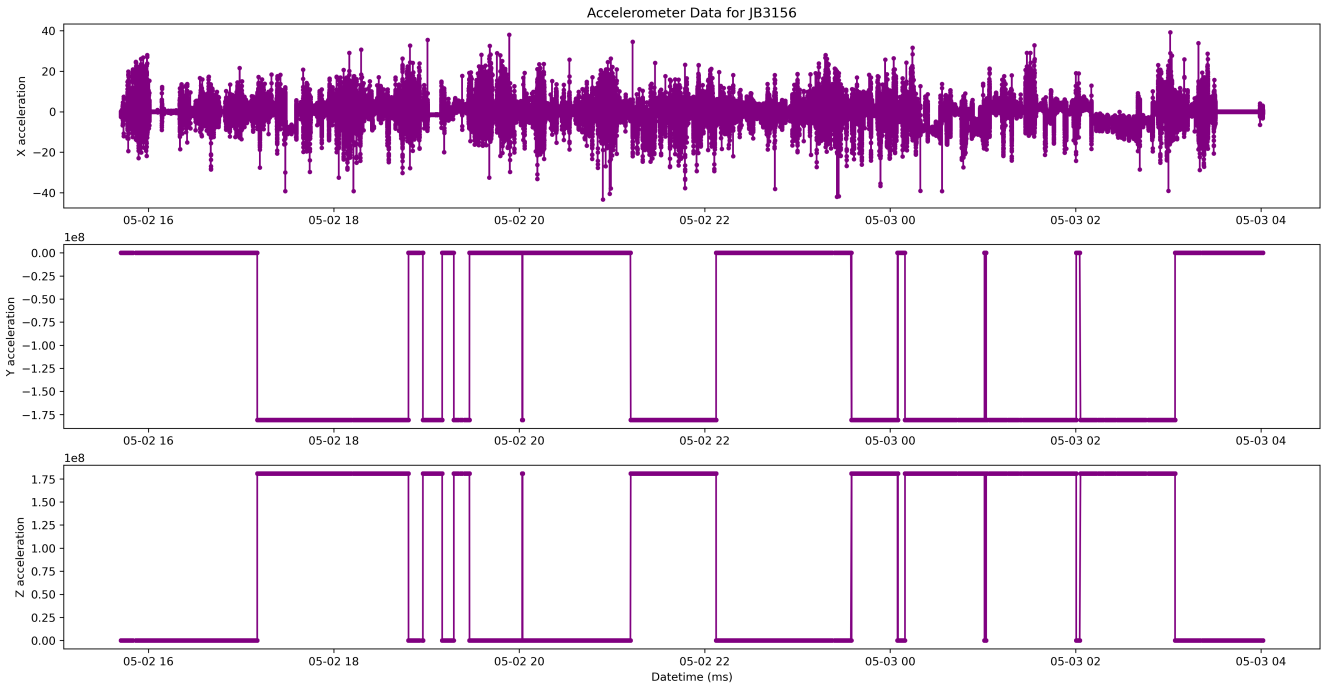


Figure 5: **Accelerometer data for one participant of the event throughout the event**

Note: Inspection of the raw accelerometer timestamps contradicts the reported uniform 40 Hz sampling rate. The X-acceleration trace shows non-uniform clustering of data points, while the Y and Z axes show prolonged periods of constant values. These patterns suggest significant data loss or buffering issues, resulting in highly irregular sampling intervals rather than a continuous 40 Hz stream.

pid	Min Interval (ms)	Median Interval (ms)	Max Interval (ms)
JB3156	0.0	20.0	114899.0
CC6740	0.0	25.0	2157503.0
SA0297	0.0	45.0	9366095.0
PC6771	0.0	25.0	21320666.0
BK7610	0.0	25.0	825778.0
DC6359	0.0	25.0	8148762.0
MC7070	0.0	25.0	15147211.0
MJ8002	0.0	25.0	12773235.0
BU4707	0.0	25.0	6588017.0
JR8022	0.0	26.0	12233634.0
HV0618	0.0	25.0	16572634.0
SF3079	0.0	26.0	25497524.0
DK3500	0.0	25.0	16849433.0

Table 2: Minimum, median, and maximum time intervals between accelerometer samples for each participant.

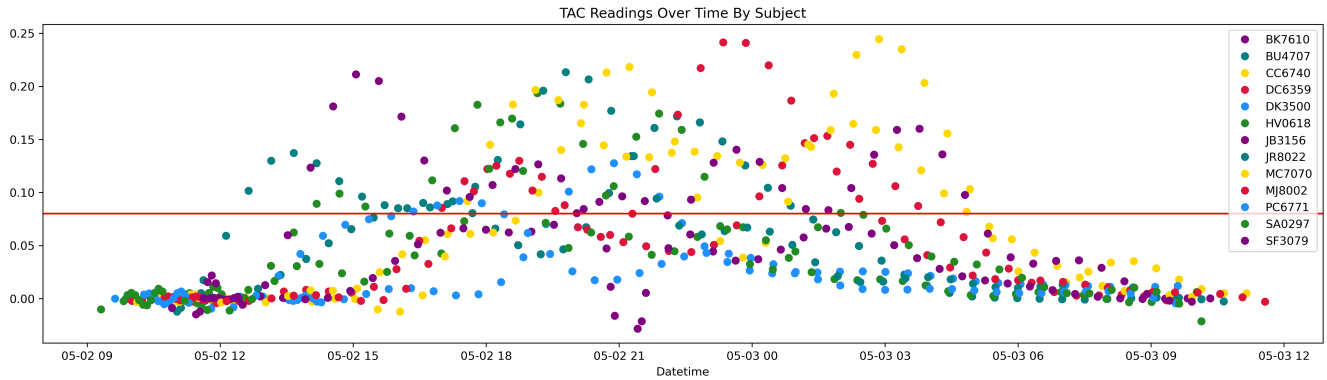


Figure 6: **TAC readings over time by participating subject**

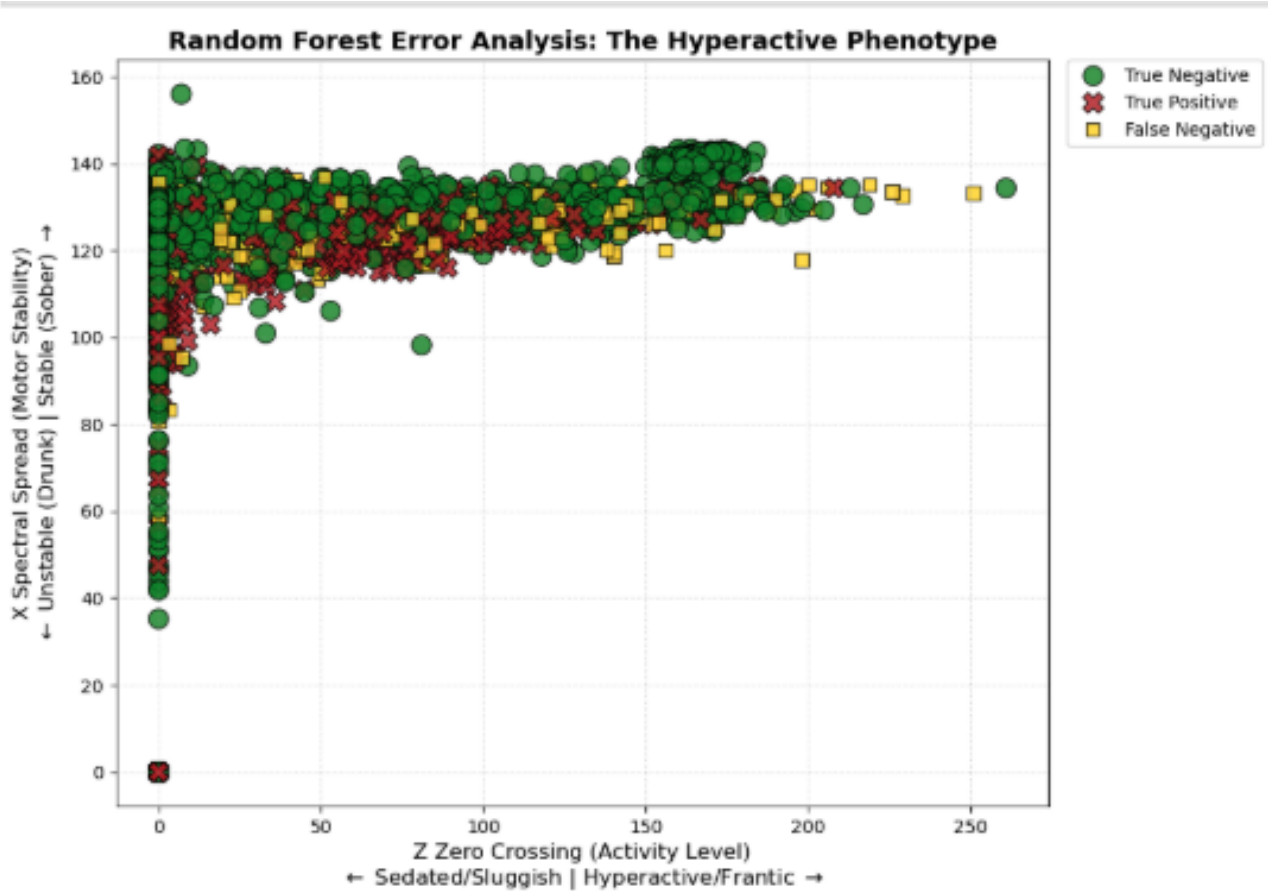
Note: The graph below shows the spread of raw TAC readings by participant, with the "intoxicated" threshold of 0.08 g/dl in red. It looks like there might be potential issues with class imbalance.

5.3 Feature Engineering

Table 3: Summary of Time and Frequency Domain Features

Feature (x,y,z)	**Domain**	**Window Size**	**Description & Meaning**
Mean	Time	1 s	Average acceleration; indicates static posture.
Median	Time	1 s	measure of central tendency; filters outliers.
Standard Deviation	Time	1 s	Signal variability; indicates intensity of movement.
Min / Max	Time	1 s	Extremes of motion; captures peaks or impacts.
Percentiles (5, 25, 75, 95)	Time	10 s	Distribution of the signal values over the full window.
Zero Crossing Rate	Time	10 s	Frequency of sign changes; correlates with movement frequency.
Total Energy	Time	10 s	Sum of squared signal values; overall vigor of activity.
Energy Entropy	Time	10 s	Randomness of energy distribution; consistency of movement.
Spectral Centroid	Frequency	10 s	Center of mass of the spectrum; distinguishes low vs. high freq. motion.
Spectral Spread	Frequency	10 s	Bandwidth of the spectrum; indicates rhythmic vs. chaotic gait.

5.4 Error Analysis



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